Airbnb NYC Price Analysis

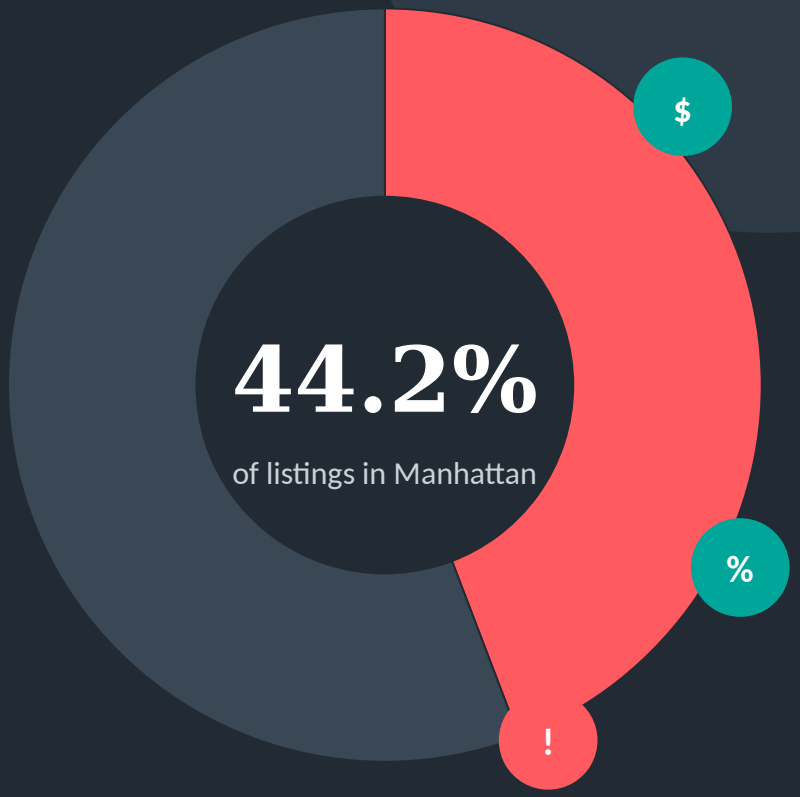
From Raw Listings to Pricing Insight - A Full EDA Journey

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48,895	16	264	$105
Raw Listings	Columns	Rows Removed	Median Price

PROJECT BRIEF

Problem Statement & Objective

! Problem Statement

- Airbnb hosts often struggle to price listings correctly
- Underpricing leaves revenue on the table; overpricing drives away bookings
- Raw listing data is messy — missing values, invalid entries, extreme outliers
- Without data-driven insight, pricing decisions rely on guesswork

> Objective

- Clean a real 48,895-listing NYC Airbnb dataset end-to-end
- Uncover true pricing drivers — location, room type, and their interaction
- Build a solid, insight-rich foundation for a data-driven price recommendation model
- Set direction for feature engineering and model selection

Dataset Overview

NYC Airbnb Listings 2019

48,895

raw listings collected

16

features (1 is the target)

48,631

rows after cleaning

Numerical Columns

price

Nightly price in US dollars (target)

minimum_nights

Minimum stay required

number_of_reviews

Total reviews received

availability_365

Days available per year

Categorical Columns

neighbourhood_group

5 boroughs; plus 221 neighbourhoods

room_type

Entire home, private or shared room

name, host_name

Listing and host text fields

last_review

Date of the latest review

Target Column: price

Nightly price in US dollars, from $0 to $10,000 in the raw data. It drives every cleaning, EDA and modeling decision in this project.

Duplicate & Missing Value Checks

Duplicates were checked first (none found), then every column with missing values. Sample affected rows from the raw data:

0
Exact duplicate rows

16
Listings with missing name

21
Listings with missing host_name

10,052
Listings with no review date

Row	id	name	host_name	neighbourhood_group	room_type	price	last_review	reviews_per_month
Row 2854	1615764	NaN	Peter	Manhattan	Entire home/apt	400	NaN	NaN
Row 3703	2232600	NaN	Anna	Manhattan	Entire home/apt	200	2015-06-08	0.45
Row 5775	4209595	NaN	Jesse	Manhattan	Entire home/apt	225	2015-01-01	0.02
Row 5975	4370230	NaN	Michaël	Manhattan	Entire home/apt	215	2016-01-02	0.09
Row 2	3647	THE VILLAGE OF HARLEM...	Elisabeth	Manhattan	Private room	150	NaN	NaN
Row 19	7750	Huge 2 BR Upper East Ce	Sing	Manhattan	Entire home/apt	190	NaN	NaN

Showing 4 of 16 listings with missing names (pink) and 2 of 10,052 listings with no review date (amber). Every missing review date belongs to a listing with zero reviews.

Column-by-Column Cleaning Summary

Column	Issue Identified	Cleaning Action
Full Dataset	0 exact duplicate rows	No action needed
name	16 missing values	Filled with "Unknown"
host_name	21 missing values	Filled with "Unknown"
price	11 listings at $0; 239 above $1,000 (max $10,000)	Removed invalid and extreme-tail rows
minimum_nights	Max 1,250 nights; 14 listings above 365	Removed rows above 365 nights
last_review	Stored as text; 10,052 missing	Converted to datetime; gaps kept (no reviews)
reviews_per_month	10,052 missing values	Filled with 0 (zero reviews)
Other 10 columns	id, host_id, boroughs, location, room_type, reviews, host listings, availability	Valid ranges: no cleaning needed

DATA CLEANING

Raw vs. Cleaned Data - Side by Side

Same listings, shown before and after the cleaning pipeline. Highlighted cells mark the exact issues fixed.

BEFORE

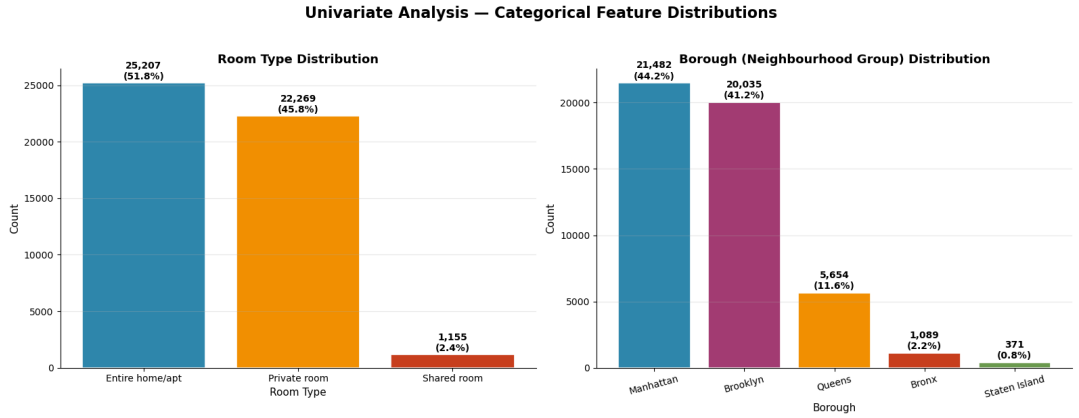
id	name	host_name	neighbourhood_group	room_type	price	minimum_nights	reviews_per_month
2232600	**NaN**	Anna	Manhattan	Entire home/apt	200	1	0.45
100184	Bienvenue	**NaN**	Queens	Private room	50	1	0.45
7750	Huge 2 BR Upper East..	Sing	Manhattan	Entire home/apt	190	7	**NaN**
18750597	Huge Brooklyn Browns..	Kimberly	Brooklyn	Private room	**0**	4	0.05
4204302	Prime W. Village loc..	Genevieve	Manhattan	Entire home/apt	180	**1250**	0.03

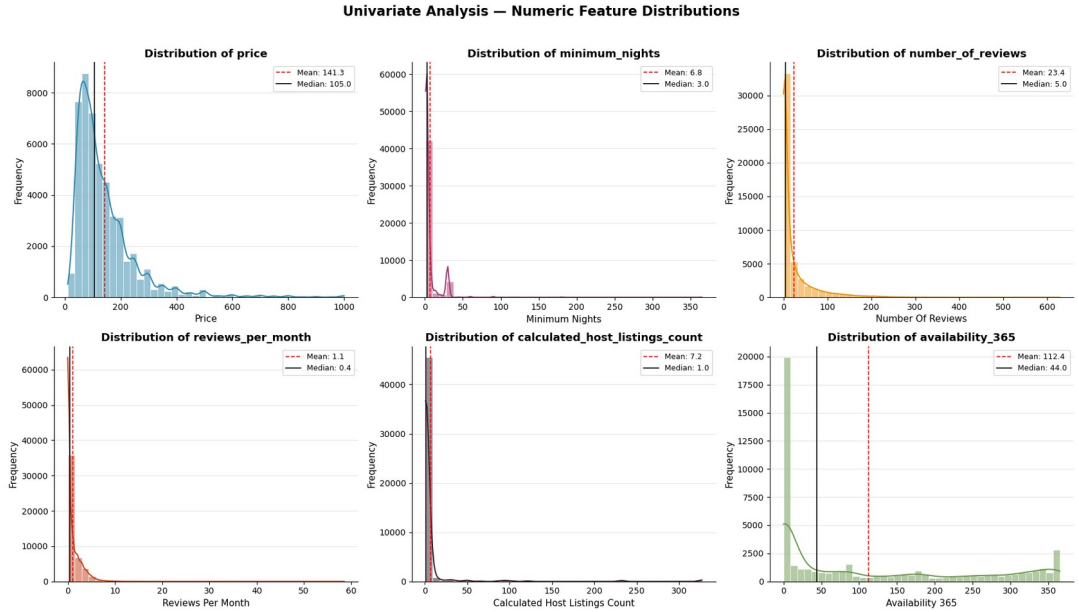
AFTER

id	name	host_name	neighbourhood_group	room_type	price	minimum_nights	reviews_per_month
2232600	Unknown	Anna	Manhattan	Entire home/apt	200	1	0.45
100184	Bienvenue	Unknown	Queens	Private room	50	1	0.45
7750	Huge 2 BR Upper East..	Sing	Manhattan	Entire home/apt	190	7	0.0

Missing names, host names and review rates are filled. The listing with a $0 price and the one with a 1,250-night minimum stay are removed, so AFTER shows 3 rows.

Univariate Analysis: Listing Mix & Distributions





What this tells us

$105

Median price

$141

Mean price

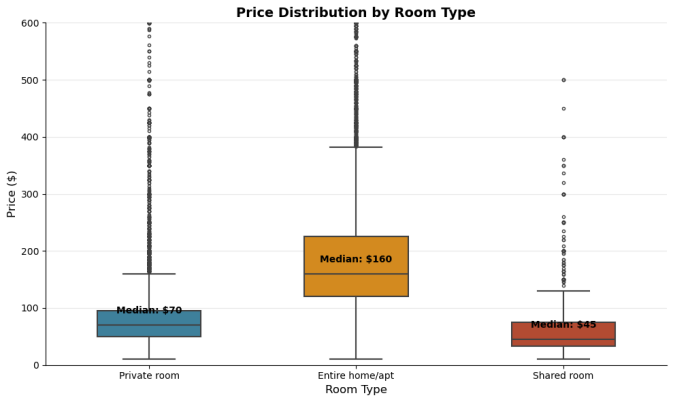
Entire homes (52%) and private rooms (46%) dominate supply, and Manhattan and Brooklyn hold 85% of listings.

Observation:

Price is right-skewed: a few luxury listings pull the mean above the median. Minimum nights, reviews and reviews per month are also strongly right-skewed, and availability spikes at 0 days (35.9% of listings).

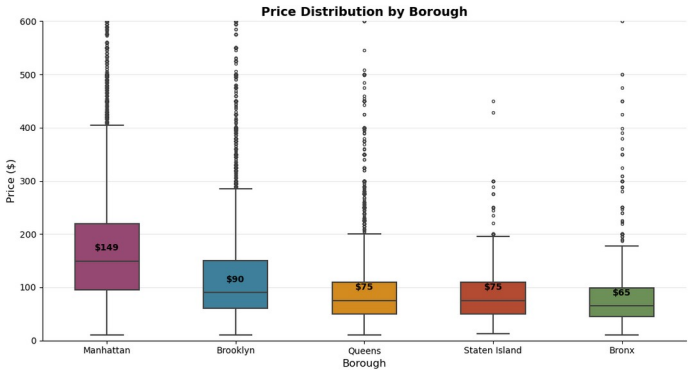
Price Drivers: Room Type & Borough

Price by Room Type



Entire home/apt	**~$160**
Private room	~$70
Shared room	~$45

Price by Borough



Manhattan	**~$149**
Brooklyn	~$90
Queens	~$75

Insight

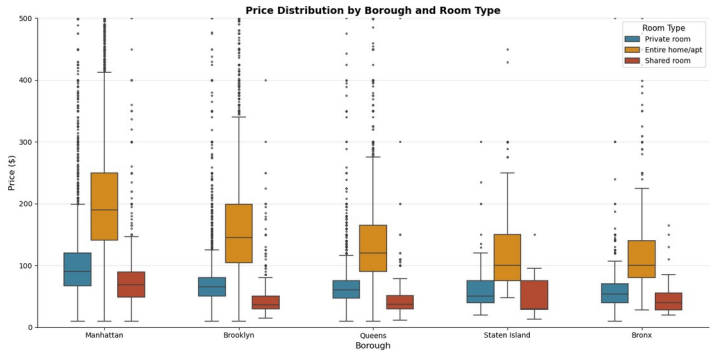
Entire homes have a median of $160, more than double a private room ($70) and over three times a shared room ($45). Room type is the strongest single price driver.

Insight

Manhattan's median price ($149) is more than double the Bronx ($65). Location is the second major driver, and Manhattan listings are priced at a clear premium.

Price by Borough x Room Type & Numeric Features

Price by Borough and Room Type

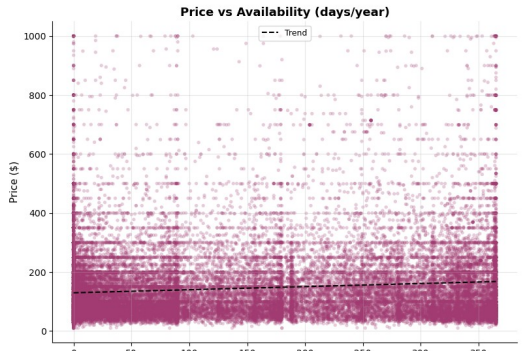


Manhattan entire home	**~$190**
Brooklyn entire home	~$145
Queens entire home	~$120
Bronx entire home	~$100

Numeric Features vs. Price



Reviews vs. Price



Availability vs. Price

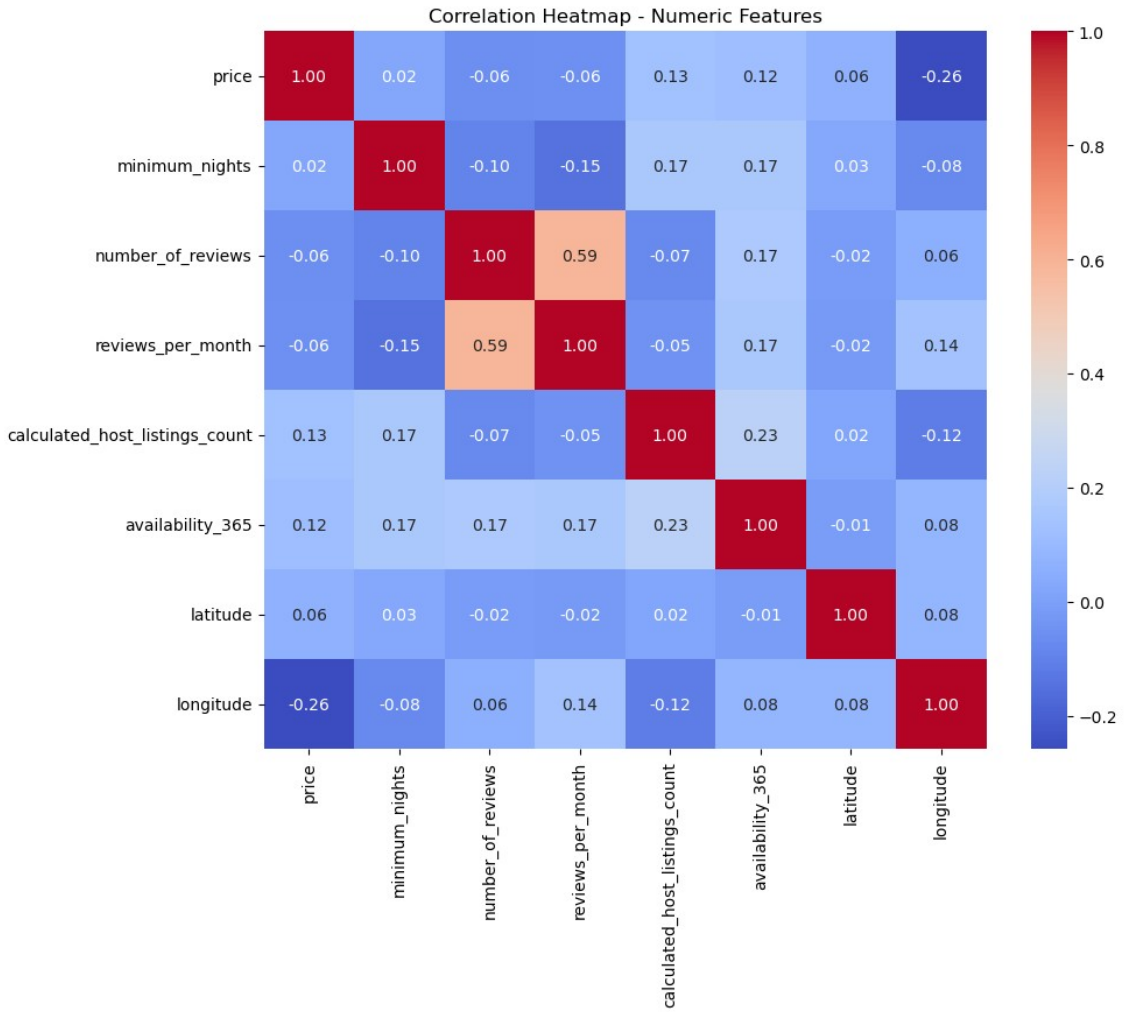
Insight

Entire homes cost the most in every borough, led by Manhattan ($190). The gap between room types is wide everywhere, so borough and room type work together.

Observations

More reviews do not mean higher prices (r = -0.06), and availability is only weakly related (r = 0.12). Numeric features alone do not explain price.

Correlation Heatmap - Numeric Features



Reading the Heatmap

Price and longitude: the strongest link (r = -0.26). Prices fall as listings move east, away from Manhattan.

number_of_reviews and reviews_per_month: moderate positive correlation (0.59), expected since both measure review activity.

All other features show weak correlation with price (r below 0.15), so no single numeric column explains it.

KEY FINDING

The Premium Price Segment

What happens when the two strongest price drivers combine?

Manhattan Borough + **Entire Home Room Type** =

$190

median nightly price in the segment

vs. $105 for all listings - about 1.8x the market median

Why this matters

- This segment holds 13,048 listings, 26.8% of the market
- Guests pay the most here, against only $30 for a Staten Island shared room
- A clear benchmark for hosts pricing premium listings: location plus room type

BUSINESS IMPACT

Market Concentration

Turning listing counts into a clear picture of where supply and pricing power sit.

44.2%

of all listings are in Manhattan

$149 median price

85.4%

of listings sit in Manhattan and Brooklyn

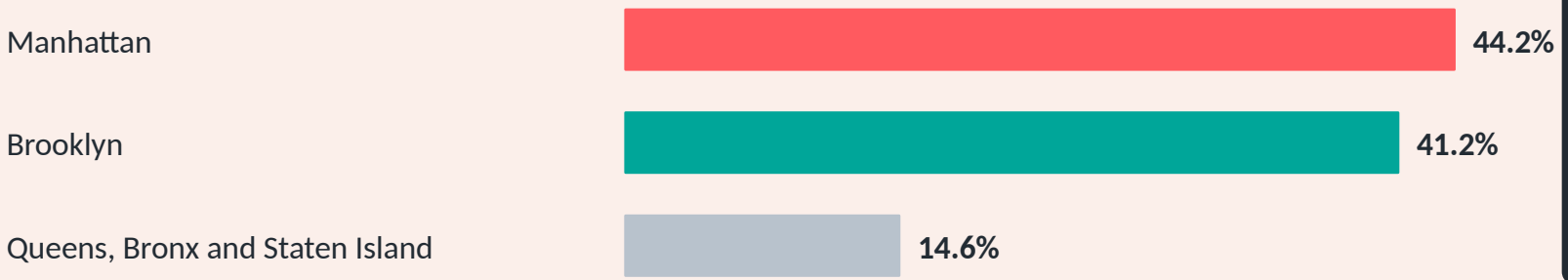
$149 and $90 median prices

35.9%

of listings are unavailable all year

17,463 listings with 0 days available

Share of Listings by Borough



85% of supply sits in two boroughs whose median prices differ by $59

SUMMARY

Key Insights from EDA

1 **Median nightly price is $105**

Mean is $141: a few luxury listings skew the distribution

2 **Entire homes cost 2.3x private rooms**

Room type is the strongest single price driver

3 **Manhattan is the priciest borough at $149**

The Bronx median is just $65: location drives a large gap

4 **Manhattan and Brooklyn hold 85% of listings**

Supply is concentrated in two boroughs

5 **Manhattan entire homes reach $190**

The premium segment: about 1.8x the overall median

6 **Numeric features correlate weakly with price**

Strongest is longitude (r = -0.26): room type and location matter more

Conclusion

What this analysis found

This EDA turned a messy 48,895-listing Airbnb dataset into a clean, machine-learning-ready dataset of 48,631 listings. It showed that price is driven mainly by room type and location, with Manhattan entire homes as the premium segment at a $190 median nightly price.

Data Cleaning • Exploratory Data Analysis • Business Insight Generation

Thank You

Questions & Discussion Welcome

$105	$190	48,631
Median nightly price	Manhattan entire homes	Clean listings

SP

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Key Insights